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**PHYSICS**

**9702/21**

Paper 2 AS Level Structured Questions

**May/June 2017**

MARK SCHEME

Maximum Mark: 60

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

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| Question   | Answer                                                                                                                                                                                      | Marks     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1(a)       | (stress =) force / area <b>or</b> $\text{kg m s}^{-2} / \text{m}^2$                                                                                                                         | <b>B1</b> |
|            | $= \text{kg m}^{-1} \text{s}^{-2}$                                                                                                                                                          | <b>A1</b> |
| 1(b)(i)    | $0.58 = 2\pi \times [(4 \times 0.500 \times 0.600^3) / (E \times 0.0300 \times 0.00500^3)]^{0.5}$                                                                                           | <b>C1</b> |
|            | $E = [4\pi^2 \times 4 \times 0.500 \times (0.600)^3] / [(0.58)^2 \times 0.0300 \times (0.00500)^3]$<br>$= 1.35 \times 10^{10} \text{ (Pa)}$                                                 | <b>C1</b> |
|            | $= 14 \text{ (13.5) GPa}$                                                                                                                                                                   | <b>A1</b> |
| 1(b)(ii)1. | (accuracy determined by) the closeness of the value(s)/measurement(s) to the true value                                                                                                     | <b>B1</b> |
|            | (precision determined by) the range of the values/measurements                                                                                                                              | <b>B1</b> |
| 1(b)(ii)2. | $l$ is (cubed so) $3 \times$ (percentage/fractional) uncertainty<br><b>and</b> $T$ is (squared so) $2 \times$ (percentage / fractional) uncertainty<br><b>and</b> (so) $l$ contributes more | <b>B1</b> |

| Question   | Answer                                                                                   | Marks     |
|------------|------------------------------------------------------------------------------------------|-----------|
| 2(a)       | resultant force (in any direction) is zero                                               | <b>B1</b> |
|            | resultant torque/moment (about any point) is zero                                        | <b>B1</b> |
| 2(b)(i)    | $a = (v - u) / t$ <b>or</b> gradient <b>or</b> $\Delta v / (\Delta)t$                    | <b>C1</b> |
|            | e.g. $a = (8.8 - 4.6) / (7.0 - 4.0) = 1.4 \text{ m s}^{-2}$                              | <b>A1</b> |
| 2(b)(ii)   | $s = 4.6 \times 4 + [(8.8 + 4.6) / 2] \times 3$                                          | <b>C1</b> |
|            | $= 18.4 + 20.1$                                                                          | <b>A1</b> |
|            | $= 39 \text{ (38.5) m}$                                                                  |           |
| 2(b)(iii)  | $\Delta E = \frac{1}{2} \times 95 [(8.8)^2 - (4.6)^2]$                                   | <b>C1</b> |
|            | $= 3678 - 1005$                                                                          | <b>A1</b> |
|            | $= 2700 \text{ (2673) J}$                                                                |           |
| 2(b)(iv)1. | weight = $95 \times 9.81$ (= 932 N)                                                      | <b>C1</b> |
|            | vertical tension force = $280 \sin 25^\circ$ <b>or</b> $280 \cos 65^\circ$ (=118.3 N)    | <b>C1</b> |
|            | $F = 932 + 118$                                                                          | <b>A1</b> |
|            | $= 1100 \text{ (1050) N}$                                                                |           |
| 2(b)(iv)2. | horizontal tension force = $280 \cos 25^\circ$ <b>or</b> $280 \sin 65^\circ$ (= 253.8 N) | <b>C1</b> |
|            | resultant force = $95 \times 1.4$ (= 133 N)                                              | <b>C1</b> |
|            | $133 = 253.8 - R$                                                                        | <b>A1</b> |
|            | $R = 120 \text{ (120.8) N}$                                                              |           |

| Question | Answer                                                                                                                                                                                            | Marks     |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)     | $\rho = m / V$                                                                                                                                                                                    | <b>C1</b> |
|          | $V = \pi d^2 L / 4$ <b>or</b> $\pi r^2 L$                                                                                                                                                         | <b>C1</b> |
|          | weight = $2.7 \times 10^3 \times \pi (1.2 \times 10^{-2})^2 \times 5.0 \times 10^{-2} \times 9.81 = 0.60 \text{ N}$                                                                               | <b>A1</b> |
| 3(b)(i)  | the point from where (all) the weight (of a body) seems to act                                                                                                                                    | <b>B1</b> |
| 3(b)(ii) | $W \times 12$                                                                                                                                                                                     | <b>C1</b> |
|          | $(0.25 \times 8) + (0.6 \times 38)$                                                                                                                                                               | <b>C1</b> |
|          | $W = (2 + 22.8) / 12$<br>$= 2.1 \text{ (2.07) N}$                                                                                                                                                 | <b>A1</b> |
| 3(c)(i)  | pressure changes with depth (in water)<br><b>or</b><br>pressure on bottom (of cylinder) different from pressure on top                                                                            | <b>B1</b> |
|          | pressure on bottom of cylinder <u>greater than</u> pressure on top<br><b>or</b><br>force (up) on bottom of cylinder <u>greater than</u> force (down) on top                                       | <b>B1</b> |
| 3(c)(ii) | anticlockwise moment reduced and reducing the weight of X reduces clockwise moment<br><b>or</b><br>anticlockwise moment reduced so clockwise moment now greater than (total) anticlockwise moment | <b>B1</b> |

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| Question | Answer                                                                | Marks     |
|----------|-----------------------------------------------------------------------|-----------|
| 4(a)     | (two) waves travelling (at same speed) in opposite directions overlap | <b>B1</b> |
|          | waves (are same type and) have same frequency/wavelength              | <b>B1</b> |
| 4(b)(i)  | $\lambda = 12 / 250 (= 0.048 \text{ m})$                              | <b>C1</b> |
|          | distance = $1.5 \times 0.048$<br>= 0.072 m                            | <b>A1</b> |
| 4(b)(ii) | $T = 1 / 250$<br>= 0.004 (s) or 4 (ms)                                | <b>C1</b> |
|          | 1. curve drawn is mirror image of that in Fig. 4.2 and labelled P     | <b>A1</b> |
|          | 2. horizontal line drawn between A and B and labelled Q               | <b>A1</b> |

| Question | Answer                                                                                     | Marks     |
|----------|--------------------------------------------------------------------------------------------|-----------|
| 5(a)     | observed frequency is different to source frequency when source moves relative to observer | <b>B1</b> |
| 5(b)     | $360 = (400 \times 340) / (340 \pm v)$                                                     | <b>C1</b> |
|          | $v = 38 (37.8) \text{ m s}^{-1}$                                                           | <b>A1</b> |
|          | away (from the observer)                                                                   | <b>B1</b> |

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| Question  | Answer                                                                                                                                      | Marks     |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)      | volt / ampere                                                                                                                               | <b>B1</b> |
| 6(b)(i)   | $R_T = [1/3.0 + 1/6.0]^{-1} + 4.0 (= 6.0 \Omega)$                                                                                           | <b>C1</b> |
|           | $I = 1.5 / 6.0$                                                                                                                             | <b>C1</b> |
|           | $= 0.25 \text{ A}$                                                                                                                          | <b>A1</b> |
| 6(b)(ii)  | $V_B = 0.5 \text{ V}$<br>$I = 0.5 / 3.0$<br>$= 0.17 (0.167) \text{ A}$                                                                      | <b>A1</b> |
| 6(b)(iii) | $P = I^2 R$ or $VI$ or $V^2/R$                                                                                                              | <b>C1</b> |
|           | ratio $= (0.167^2 \times 3.0) / (0.25^2 \times 4.0)$<br>$= 0.33$                                                                            | <b>A1</b> |
| 6(c)(i)   | vary/change/different radius/diameter/ <u>cross-sectional</u> area (of wire)                                                                | <b>B1</b> |
| 6(c)(ii)  | $v = I / Ane$<br>ratio $= \frac{(I_B / A_B)}{(I_C / A_C)}$ or $\frac{I_B}{I_C} \times \frac{A_C}{A_B}$                                      | <b>C1</b> |
|           | $(R \propto 1/A \text{ so}) \text{ ratio} = \frac{I_B}{I_C} \times \frac{R_B}{R_C} = \frac{0.167}{0.25} \times \frac{3.0}{4.0}$<br>$= 0.50$ | <b>A1</b> |
| 6(d)(i)   | 0.25 A to 0.13 (0.125) A or halved                                                                                                          | <b>A1</b> |
| 6(d)(ii)  | no change                                                                                                                                   | <b>A1</b> |

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| Question       | Answer                                                                                                                                                                                                                                             | Marks     |           |           |                |    |    |               |    |    |           |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|----------------|----|----|---------------|----|----|-----------|
| 7(a)(i)        | (proton is uud so) $(2/3)e + (2/3)e - (1/3)e = e$                                                                                                                                                                                                  | <b>B1</b> |           |           |                |    |    |               |    |    |           |
| 7(a)(ii)       | (neutron is udd so) $(2/3)e - (1/3)e - (1/3)e = 0$                                                                                                                                                                                                 | <b>B1</b> |           |           |                |    |    |               |    |    |           |
| 7(b)(i)        | <table border="1"> <tr> <td></td><td><math>\beta^-</math></td><td><math>\beta^+</math></td></tr> <tr> <td>nucleon number</td><td>90</td><td>64</td></tr> <tr> <td>proton number</td><td>39</td><td>28</td></tr> </table> <p><i>all correct</i></p> |           | $\beta^-$ | $\beta^+$ | nucleon number | 90 | 64 | proton number | 39 | 28 | <b>B1</b> |
|                | $\beta^-$                                                                                                                                                                                                                                          | $\beta^+$ |           |           |                |    |    |               |    |    |           |
| nucleon number | 90                                                                                                                                                                                                                                                 | 64        |           |           |                |    |    |               |    |    |           |
| proton number  | 39                                                                                                                                                                                                                                                 | 28        |           |           |                |    |    |               |    |    |           |
| 7(b)(ii)       | weak (nuclear force/interaction)                                                                                                                                                                                                                   | <b>B1</b> |           |           |                |    |    |               |    |    |           |
| 7(b)(iii)      | <p><math>\beta^-</math> decay: electron and (electron) antineutrino<br/> <math>\beta^+</math> decay: positron and (electron) neutrino<br/> <i>all correct</i></p>                                                                                  | <b>B1</b> |           |           |                |    |    |               |    |    |           |